# Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

## Summary

Software Engineer with 2+ years of experience building **distributed systems**, **event-driven architectures**, and **Python/C++ backend services**. Strong foundation in **quantitative problem-solving**, **testing**, and performance-critical systems.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

## Experience

**USC Dornsife** **Los Angeles, CA**
*Software Engineer* *September 2025 – Present*

- Architected a **fault-tolerant async orchestration system** using FastAPI background workers, processing 1K+ jobs with explicit **state transitions** and failure isolation, improving throughput by **3x** under concurrent load.
- Designed **ETL-style data integration pipelines** across PostgreSQL and SQL Server with incremental loads, reconciliation checks, and **automated validation**, achieving **99%+ reliability** across 5K+ monthly production jobs.

**Grad8** **Los Angeles, CA**
*Software Engineer Intern* *June 2025 – August 2025*

- Engineered a **concurrent batch-processing pipeline** using **Python** multithreading and secure file ingestion via AWS RDS and REST APIs, processing 1K+ files per batch with **fault-tolerant execution** and cost tracking.
- Built production backend on AWS serving 150+ users handling **10K+ daily requests**; implemented CI/CD-driven workflows with **automated testing**, reducing feedback turnaround by **60%** under peak load conditions.

**USC AD Labs** **Los Angeles, CA**
*ML Research Assistant* *January 2025 – May 2025*

- Built a GPT-based Decision Transformer for offline RL in autonomous driving using **Python** and **PyTorch**, developing end-to-end **data pipelines** processing 100K+ trajectory logs for reproducible experimentation.
- Architected a **spatial-temporal encoder** integrating HD-map geometry with ego-agent context via **multi-head attention**, improving closed-loop planning stability by **15%** through rigorous **test-driven** evaluation.

**Aasma Technology Solutions** **India**
*Software Developer Engineer – (SecureSync)* *August 2023 – July 2024*

- Engineered cross-platform **microservice** architecture with automated VM testing, executable builds, and 20+ **unit/integration test suites** using Jenkins, **PyTest**, and Mockito in **Agile Scrum** with Microsoft/GitHub stakeholders.
- Optimized large-scale API integrations using **OOP design patterns**, AWS Glue, and S3, improving response times by **30%**. Led **ETL workflows** boosting pipeline efficiency **68%** and cutting technical debt by 20%.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern – (VibrantChem)* *May 2023 – August 2023*

- Developed MERN-based platform with **WebSockets** and RESTful APIs supporting **event-driven** real-time updates across 8 **microservices**, syncing 2M+ records across locations with 33% fewer sync conflicts.
- Automated **Docker-based build and test flows** with Jenkins CI/CD, cutting release time by **50%**, improving deployment consistency and reducing environment drift across **distributed** staging and production environments.

## Skills

**Languages**: Python, C++, Java, JavaScript/TypeScript, SQL, Shell, Bash
**Backend & APIs**: FastAPI, Flask, Node.js, Express.js, REST APIs, Spring Boot, Django
**Distributed & Event-Driven**: Microservices, Event-Driven Architecture, Async Pipelines, WebSockets, Message Queues
**Databases**: PostgreSQL, MySQL, SQL Server, MongoDB, Redis, Firebase
**Cloud & Infrastructure**: AWS (EC2, S3, RDS, Glue), Docker, Kubernetes, NGINX, CI/CD, Jenkins
**Testing & Practices**: TDD, PyTest, Mockito, Refactoring, Unit/Integration Testing, Agile/Scrum, Git
**AI/ML**: PyTorch, TensorFlow, Transformers, Feature Engineering, Hyperparameter Tuning

## Projects

**CPU-Scheduling Simulator** Live | *C++, React.js, JavaScript, OS Internals*

- Built a **C++ scheduling engine** implementing FCFS, SJF, SRT, RR, and MLQ algorithms with **performance optimization**, reducing average turnaround time by **25%** across 1K+ simulated process combinations.

**Chat2SQL** Code | *Java, TypeScript, SQL, Docker, Maven*

- Designed a schema-aware **SQL transformation pipeline** generating optimized queries with 82% accuracy, implementing **validation and structured execution** against relational databases across a **Java/TypeScript** codebase.

**Linguistix** Code | *Python, AWS, Docker, CI/CD*

- Deployed a containerized **Python** RESTful data-processing API on AWS ECS/Fargate with **automated CI/CD** validation gates, achieving 87% accuracy and **production-grade reliability** across deployment environments.